

Name _____ Period _____

Skill 29.1 Exercise 1

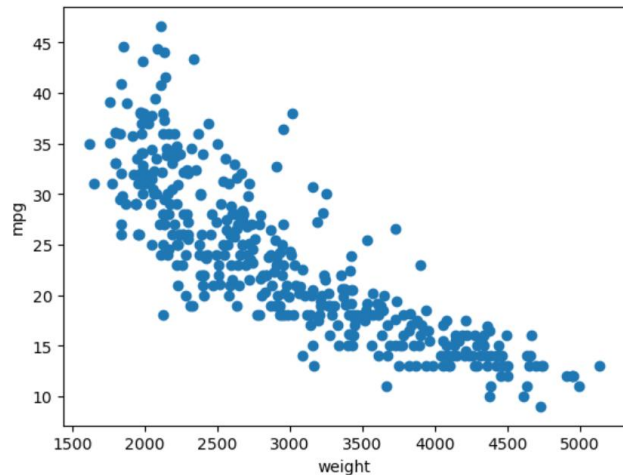
Refer to the `cars` dataframe below,

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	car name
0	18.0	8	307.0	130	3504	12.0	70	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	ford torino

Complete the spaces below to make a scatter plot to see if there is an association between `mpg` and `weight`.

```
plt.scatter(x = , y = )  
plt.xlabel(' ')  
plt.ylabel(' ')  
plt.show()
```

Below is the actual plot



Based on the scatter plot, does vehicle weight appear to be associated with fuel efficiency (mpg)? Describe the direction and strength of the association.

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Skill 29.2 Exercise 1

Refer to the cars DataFrame below.

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	car name
0	18.0	8	307.0	130	3504	12.0	70	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	ford torino

Use the `cov()` method to calculate the covariance matrix for *mpg* and *weight*.

The actual matrix is as follows,

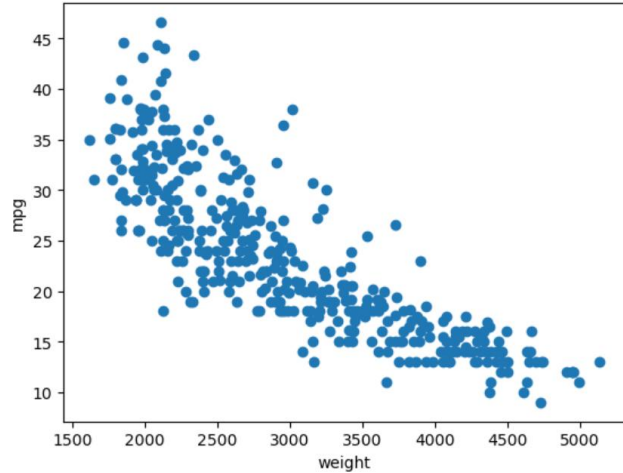
```
           weight      mpg
weight  717140.990526 -5505.211745
mpg      -5505.211745   61.089611
```

Identify which value in the matrix represents the covariance between weight and mpg. What does its sign tell you about their relationship?

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Skill 29.3 Exercise 1

Below is the plot of the *mpg* and *weight*.



Use the *pearsonr* function from *scipy.stats* to calculate the correlation between *mpg* and *weight* and print the result.

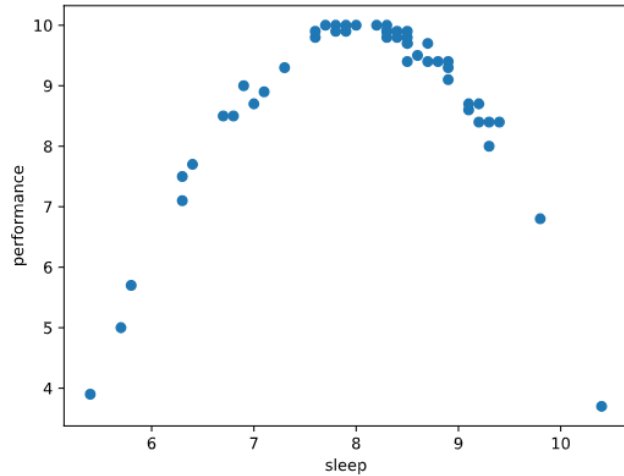
```
from scipy.stats import pearsonr
```

The actual value is -0.831740933244335. Is this value consistent with the scatter plot?

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Skill 29.4 Exercise 1

The plot below shows the relationship between hours of sleep and performance score on a task.



Describe the relationship between sleep and performance shown in the scatter plot.

Would correlation accurately reflect the strength of this relationship?

If the correlation were near zero, would that mean sleep and performance are unrelated? Why or why not?